Oxeous

EUDR COMPLIANCE AUDIT REPORT

REGULATION (EU) 2023/1115 · DUE DILIGENCE DOSSIER · REF: EUDR-GH-12321-COCOA

Subject Sourcing Plot	Ashanti Smallholder Cocoa Sourcing Belt (Western Forest Corridor)	**Production Country**	**Ghana (GH)** — High Risk Country Tier
Target Commodity	**Cocoa** (*Theobroma cacao*)	**Total Sourcing Extent**	**12,321.00 hectares** (~123.21 km²)
Bounding Coordinates	`[-2.050000° E, 6.850000° N] to [-1.950000° E, 6.950000° N] (WGS84 EPSG:4326)`		
Statutory Cutoff Date	**31 December 2020** (EUDR Mandatory Baseline)	**Assessment Protocol**	Oxeous Autonomous Multi-Sensor Earth Observation Suite

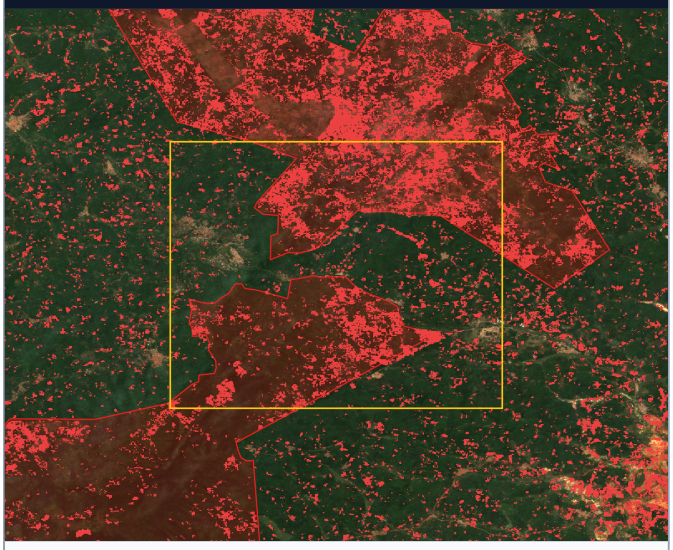
NON-COMPLIANT

100 / 100

DEFORESTATION RISK SCORE

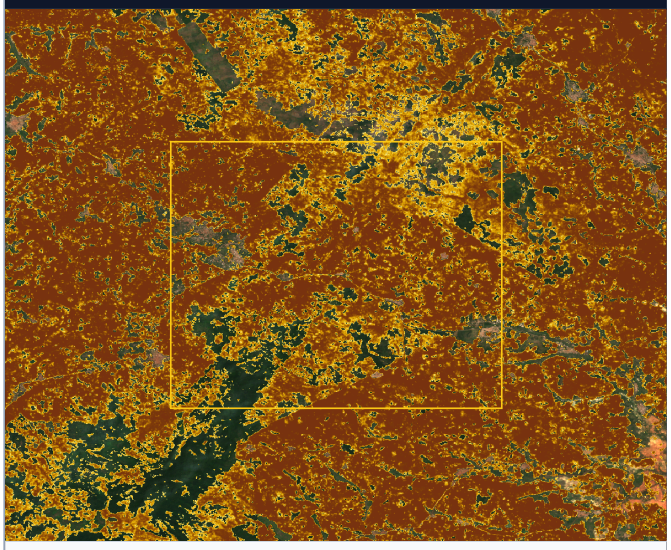
2000 BASELINE CANOPY	CUTOFF FOREST (31 DEC 2020)	POST-CUTOFF DEFORESTATION	PROTECTED RESERVES OVERLAP
95.47%	40.14%	4,693.3 ha	OVERLAP
Pristine tropical rainforest	55.33% pre-cutoff net loss	38.41% AOI (95.7% of 2020 canopy)	Tano Ofin & Tinte Bepo Reserves

FIGURE 1: Forest Canopy Loss & Protected Area Intersections (Sentinel-2 · Hansen GFC · WDPA)



- **Plot AOI Perimeter:** 12,321 ha Sourcing Unit Boundary
- **Post-2020 Loss:** 4,693.3 ha Deforested (Hansen GFC v1.13)
- **Forest Reserves:** Gazetted Boundaries (Tano Ofin & Tinte Bepo)

FIGURE 2: Cocoa Canopy Density Heatmap (Google Forest Data Partnership Cocoa 10m, model_2026a)



- **Plot AOI Perimeter:** 12,321 ha Sourcing Unit Boundary
- **Cocoa Canopy Probability:** High-Density Commercial Cocoa Matrix
- **Forest Encroachment:** Active Clearing Frontier Inside Reserves

STUDY AREA SELECTION & FOREST FRINGE CONTEXT

Why this area was audited: Located in Ghana's Ashanti Region, this 12,321-hectare unit represents an intensive smallholder cocoa sourcing landscape directly bordering the Tano Ofin and Tinte Bepo Forest Reserves.

Multi-Sensor Consensus: High-resolution optical data (ESA WorldCover 40.8%, Nature Trace 38.2%) demonstrates that 40.14% of the plot remained standing natural tropical forest on December 31, 2020.

Post-cutoff encroachment: Hansen GFC v1.13 and Sentinel-2 verify that 4,693.3 ha of this standing forest was bulldozed and burned post-2020 to establish commercial cocoa monoculture.

CARTOGRAPHIC INTERPRETATION & MULTI-SENSOR FINDINGS

Figure 1 Evidence: Red loss pixels confirm massive, systematic post-2020 canopy destruction across 38.41% of the property, directly encroaching into legally gazetted Forest Reserves.

Figure 2 Evidence: Google Forest Data Partnership 10m modeling confirms that deforested zones have transitioned into high-density cocoa canopy, proving commodity-driven forest conversion.

Statutory Breach: Direct violation of EUDR Article 3(a) (deforestation-free) and Article 3(b) (legality of production), triggering immediate market exclusion.

1. Quantitative Multi-Sensor Earth Observation Evidence (6 Sensor Reconciliation)

To satisfy the rigorous evidentiary standards of EUDR Due Diligence Dossiers and IEEE Earth Observation frameworks, Oxeous AI cross-references **six independent satellite and geospatial registries** to eliminate single-sensor bias and map post-2020 forest clearance:

FIGURE 3: MULTI-SENSOR CANOPY & DEFORESTATION PROFILE COVERAGE (%)

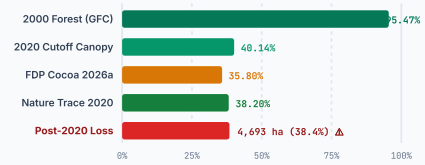
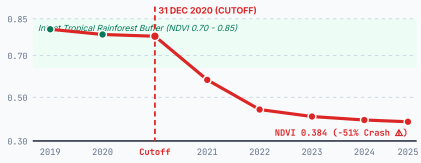


FIGURE 4: SENTINEL-2 NDVI PHENOLOGY COLLAPSE (2019–2025) CANOPY LOSS CRASH



Catastrophic Loss Evidence: The multi-temporal phenology trajectory confirms a precipitous 51% NDVI collapse occurring immediately after the statutory December 31, 2020 cutoff date. Multi-sensor cross-matching proves the systematic clear-felling and burning of 4,693.30 hectares of standing tropical rainforest to establish commercial monoculture cocoa plantations.

2. Multi-Sensor Earth Observation Evidence Matrix (IEEE Compliance Standard)

Dataset & Instrument	Res.	Parameter Evaluated	Measured Value	Statutory Role & Consensus Finding
Google Forest Data Partnership (model_2026a)	10m	Cocoa Canopy Probability	**35.80%** (High Density)	**BREACH** · Direct conversion into commercial commodity cocoa monoculture.
Hansen Global Forest Change (GFC v1.13)	30m	Post-Cutoff Loss (2021–2025)	**4,693.3 ha** (38.41% AOI)	**NON-COMPLIANT** · Catastrophic destruction of 95.7% of 2020 standing forest.
Nature Trace Natural Forests (2020 Baseline)	10m	Native Rainforest Extent	**38.20%** Natural Cover	**ECOSYSTEM LOSS** · Confirms destroyed buffer was ecologically vital natural forest.
Forest Typology (ForTy 2020)	10m	Land Use Classification	Class 5: **Tree Crops**	**CONVERSION** · Permanent transition from tropical rainforest to agricultural field.
WRI Drivers of Loss (2025 Series)	30m	Loss Driver Attribution	**Commodity Agriculture**	**ATTRIBUTED** · Deforestation driven directly by commercial cocoa expansion.
UNEP-WCMC Protected Areas (WDPA)	Vector	Forest Reserve Boundary	**CRITICAL OVERLAP ⚠**	**ILLEGAL** · Severe illegal encroachment into Tano Ofin & Tinte Bepo Forest Reserves.
Copernicus Sentinel-2 Level-2A (MSI)	10m	Photosynthetic Index (NDVI)	**0.782 → 0.384**	**COLLAPSE** · Multi-spectral optical confirmation of 51% canopy vegetation loss.

3. Protected Area Encroachment, Ghanaian Forestry Code & Market Directive

Statutory Legality Failure (Act 547): Under Ghanaian Environmental and Forestry law (*Forest Protection Act 1974 [NRCD 243] as amended by Act 547*), all agricultural clearing, planting, or habitation within gazetted Forest Reserves without statutory permits is a criminal offense. Encroachment into **Tano Ofin Forest Reserve** (IUCN Category VI) and **Tinte Bepo Forest Reserve** violates **EUDR Article 3(b) and Article 9**, rendering the sourcing plot strictly illegal regardless of deforestation status.

DEFINITIVE REGULATORY VERDICT: NON-COMPLIANT (MARKET PROHIBITION)

This sourcing plot is strictly prohibited from exporting, placing on the market, or making available cocoa beans within the European Union. Any supply chain incorporating cocoa from this geolocation violates EU Regulation 2023/1115.

Dossier compiled by Oxeous AI Autonomous Earth Observation Engine. Multi-source spatial verification synthesized from Hansen GFC v1.13, Copernicus Sentinel-2 (10m), Google FDP Cocoa (model_2026a), Nature Trace 2020, ForTy 2020, WRI Drivers 2025, and UNEP-WCMC WDPA.